SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:

I __D. Sindhu (192424197)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

D. Sindhu.

Signature of the student:

Annexure A

Short Answer Test

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 Marks)

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors. (5 Marks)

4.A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms. (5 Marks)

5 .During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model. (5 Marks)

6. A perceptron model is used to classify whether a loan is approved based on two features: income and credit score. The weights are 0.7 and 0.5, and the bias is –0.6. For an applicant with income = 1.5 and credit score = 2, compute the net input and

apply the step activation function. Determine whether the loan is approved or rejected. (5 Marks)

7. In a neural network, a single neuron uses the delta learning rule. The initial weight is 0.4, learning rate is 0.2, and input is 3. The actual output is 0.5, while the target output is 0.9. Calculate the error, determine the weight update, and compute the new weight after one iteration. (5 Marks)

8. A classification model produces the following confusion matrix values: True Positives = 120, True Negatives = 200, False Positives = 30, and False Negatives = 50. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on these results, evaluate whether the model is suitable for applications where false negatives are critical. (5 Marks)

9. A Genetic Algorithm is used to maximize the function f(x) = 2x + 3 for an initial population of values {1, 3, 5, 7}. Compute the fitness of each individual and select the top two candidates. Perform a single-point crossover to generate offspring and explain how mutation can help avoid premature convergence. (5 Marks)

10. During training of a neural network, the training loss decreases continuously, while validation loss decreases initially and then starts increasing after several epochs. Identify the issue occurring in the model. Explain the reason behind this behavior and suggest two techniques such as regularization or early stopping to improve generalization performance. (5 Marks)

11. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is –0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not. (5 Marks)

12. In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration. (5 Marks)

13. A classification model produces the following confusion matrix values: True Positives = 140, True Negatives = 180, False Positives = 25, and False Negatives = 35. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on the results, analyze whether the model is suitable for applications such as fraud detection where false positives are costly. (5 Marks)

14. A Genetic Algorithm is applied to maximize the function f(x) = 3x + 2 for an initial population {2, 5, 7, 9}. Compute the fitness value for each individual and select the best candidates. Perform a crossover operation to generate new offspring and explain how mutation helps in maintaining diversity and avoiding local optima. (5 Marks)

15. During the training of a deep learning model, the training accuracy increases from 65% to 97%, while validation accuracy increases from 63% to 75% and then fluctuates around 72%. Identify the issue present in the model. Explain why this happens and suggest techniques such as dropout and data augmentation to improve the model's performance. Also, discuss the importance of validation metrics in model evaluation. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Perceptron – Student Pass/Fail

Given:

- $w_1 = 0.6$, $w_2 = 0.4$
- Bias $b = -0.5$
- $x_1 = 2$, $x_2 = 1$

Step 1: Net Input

$$Net = (w_1 x_1) + (w_2 x_2) + b$$
$$= (0.6 \times 2) + (0.4 \times 1) - 0.5$$
$$= 1.2 + 0.4 - 0.5 = 1.1$$

Step 2: Step Activation

$$f(x) = \begin{cases} 1, & Net \geq 0 \\ 0, & Net < 0 \end{cases}$$

Since 1.1 > 0

Output = 1

Conclusion

Student Passes.

2. Backpropagation (Delta Rule)

Given

- Weight = 0.5
- Learning rate $\eta = 0.1$
- Input $x = 2$
- Target $T = 1$
- Actual Output $O = 0.6$

Error

$$Error = T - O = 1 - 0.6 = 0.4$$

Weight Update

$$\Delta w = \eta \times Error \times x$$
$$= 0.1 \times 0.4 \times 2 = 0.08$$

Updated Weight

$$w_{new} = 0.5 + 0.08 = 0.58$$

Answer

- Error = **0.4**
- Weight Update = **0.08**
- New Weight = **0.58**

3. Confusion Matrix

Given

- TP =180
- TN =150
- FP =20
- FN =50

Accuracy

$$= \frac{TP + TN}{TP + TN + FP + FN} = \frac{330}{400} = 0.825$$

Accuracy = 82.5%

Precision

$$= \frac{180}{180 + 20} = \frac{180}{200} = 90\%$$

Recall (Sensitivity)

$$= \frac{180}{180 + 50} = \frac{180}{230} = 78.26\%$$

Specificity

$$= \frac{150}{150 + 20} = \frac{150}{170} = 88.24\%$$

Conclusion

- Accuracy = **82.5%**
- Precision = **90%**
- Recall = **78.26%**
- Specificity = **88.24%**

The model is reasonably reliable, but **recall should be improved** since false negatives are important in medical diagnosis.

4. Genetic Algorithm

Function

$$f(x) = x^2$$

Population = {2,4,6,8}

Fitness

Individual	Fitness
2	4
4	16
6	36
8	64

Best Candidates

8 and 6

Binary

8 = 1000

6 = 0110

Exchange Least Significant Bit

Offspring:

1000 → **1000 (8)**

0110 → **0110 (6)**

(No change because LSBs are both 0.)

Mutation

Mutation randomly changes one bit to:

- increase diversity
- avoid local optimum
- prevent premature convergence.

5. Training vs Validation Accuracy

Training Accuracy:

60% →95%

Validation Accuracy:

58% →72% →68%

Issue

Overfitting

Reason

The model memorizes training data and cannot generalize to unseen data.

Remedies

- Dropout
- Early Stopping

(Regularization/Data Augmentation are also acceptable.)

Importance of Validation Accuracy

Validation accuracy measures the model's ability to perform on unseen data and helps detect overfitting.

6. Perceptron – Loan Approval

Given

- $w_1 = 0.7$, $w_2 = 0.5$
- $b = -0.6$
- $x_1 = 1.5$, $x_2 = 2$

Net Input

$$= (0.7 \times 1.5) + (0.5 \times 2) - 0.6$$
$$= 1.05 + 1 - 0.6 = 1.45$$

Since **1.45>0**

Output

1

Conclusion

Loan Approved

7. Delta Rule

Given

- Weight=0.4
- Learning rate=0.2
- Input=3
- Target=0.9
- Output=0.5

Error

$$0.9 - 0.5 = 0.4$$

Weight Update

$$0.2 \times 0.4 \times 3 = 0.24$$

New Weight

$$0.4 + 0.24 = 0.64$$

Answer

- Error = **0.4**

- Weight Update = **0.24**
- New Weight = **0.64**

8. Confusion Matrix

Given

- TP=120
- TN=200
- FP=30
- FN=50

Accuracy

$$= \frac{320}{400} = 80\%$$

Precision

$$= \frac{120}{150} = 80\%$$

Recall

$$= \frac{120}{170} = 70.59\%$$

Specificity

$$= \frac{200}{230} = 86.96\%$$

F1-score

$$= \frac{2PR}{P + R}$$

$$= \frac{2(0.8)(0.7059)}{0.8 + 0.7059} = 0.75$$

F1 =75%

Conclusion

Recall is only **70.59%**, so it is **not ideal where false negatives are critical**.

9. Genetic Algorithm

Function

$$f(x) = 2x + 3$$

Population

{1,3,5,7}

Fitness

x	Fitness
1	5
3	9
5	13
7	17

Best Individuals

7 and 5

Binary

7=111

5=101

Single-point crossover

111

101

↓

101 (5)

111 (7)

Mutation

Mutation changes random bits to:

- maintain diversity
- avoid premature convergence
- explore better solutions.

10. Training Loss vs Validation Loss

Training loss decreases continuously.

Validation loss decreases initially then increases.

Issue

Overfitting

Reason

The model starts memorizing training data instead of learning general patterns.

Solutions

- Early Stopping
- Regularization (L1/L2)

Benefit

These improve generalization and reduce overfitting.

11. Perceptron – Product Purchase

Given

- Weights =0.5,0.3
- Bias=-0.4
- Inputs=2,1

Net Input

$$= (0.5 \times 2) + (0.3 \times 1) - 0.4$$
$$= 1 + 0.3 - 0.4 = 0.9$$

Since positive,

Output

1

Conclusion

Customer Purchases Product

12. Delta Rule

Given

- Weight=0.6
- Learning rate=0.15
- Input=2.5
- Target=1
- Output=0.7

Error

$$1 - 0.7 = 0.3$$

Weight Update

$$0.15 \times 0.3 \times 2.5 = 0.1125$$

New Weight

$$0.6 + 0.1125 = 0.7125$$

Answer

- Error = **0.3**
- Weight Update = **0.1125**
- New Weight = **0.7125**

13. Confusion Matrix

Given

- TP=140
- TN=180
- FP=25
- FN=35

Accuracy

$$= \frac{320}{380} = 84.21\%$$

Precision

$$= \frac{140}{165} = 84.85\%$$

Recall

$$= \frac{140}{175} = 80\%$$

Specificity

$$= \frac{180}{205} = 87.80\%$$

F1-score

$$= \frac{2(0.8485)(0.8)}{0.8485 + 0.8} = 82.35\%$$

Conclusion

The model is suitable for fraud detection because it has **high precision (84.85%)**, reducing false positives.

14. Genetic Algorithm

Function

$$f(x) = 3x + 2$$

Population

{2,5,7,9}

Fitness

x	**Fitness**
2	8
5	17
7	23
9	29

Best Candidates

9 and 7

Binary

9 =1001

7 =0111

Example single-point crossover

1001

0111

↓

1011 (11)

0101 (5)

Mutation

Mutation:

- Introduces new genes,
- Maintains diversity,

- Avoids local optimum,
- Improves search capability.

15. Training vs Validation Accuracy

Training Accuracy

65% →97%

Validation Accuracy

63% →75% →72%

Issue

Overfitting

Reason

The model learns training data too well and performs poorly on unseen data.

Remedies

- Dropout
- Data Augmentation

Importance of Validation Metrics

Validation metrics evaluate the model on unseen data, help detect overfitting, compare models, and guide the selection of the best-performing model.